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Build what next India runs on

Team Name : The-Data-AI-Challenge

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Problem Statement : Intelligent Candidate Discovery & Ranking (Track: Data & AI Challenge)

Solution Overview

- **Our solution is a deterministic, multi-signal ranking system engineered for maximum speed, explainability, and compliance:**

• Deterministic Scoring & Filtering Engine: A memory-safe pipeline running completely offline on CPU in under 15 seconds, scalable to hundreds of thousands of candidates without costly LLM APIs or GPU resource constraints.

• Honeypot / Anomaly Detection Layer: Proactively screens profiles for chronological and logical inconsistencies, eliminating exactly 67 impossible fake candidates and achieving a 0.0% honeypot rate.

• Multi-Dimensional Matrix: Evaluates base candidate relevance across years of experience, current title and domain fit, product vs. consulting company history, and technical skill matrices.

• Behavioral Signal Modifiers: Multiplies base matching scores dynamically based on notice periods, response rates, recent log-in history, and interview completion rate envelopes.

JD Understanding & Candidate Evaluation

- **Deep Job Description (JD) Understanding & Multi-Signal Candidate Evaluation:**

• Experience Sweet Spot: Focuses on the target 5-9 years experience band, with maximum weighting assigned to 6-8 years total experience.

• Consulting & Service Company Penalty: Identifies and penalizes candidates whose entire histories are at services firms (TCS, Infosys, Wipro, Accenture, Capgemini) to prioritize product-engineering mindsets.

• Title & Domain Match: Scores profiles for core Information Retrieval (IR), search, ranking, RAG, and NLP experience. Unwanted disciplines like computer vision, speech, robotics, operations, or marketing are penalized.

• Availability Modifiers: Down-weights candidates who are inactive (>90 days), unresponsive to recruiter messages (low response rates), or require a >90 days notice buyout, matching real-world hireability.

Ranking Methodology

- **Mathematical Scoring Matrix & Tie-breaking workflow:**

• Pipeline Step 1: Pre-process the candidate pool line-by-line using standard Python json parsing (zero memory overhead).

• Pipeline Step 2: Honeypot Screening flags and discards invalid candidate profiles.

• Pipeline Step 3: Base Scoring (Max 80 points) evaluates Experience (10 pts), Stability & Tenure (10 pts), Title/Headline Domain Fit (15 pts), Skills (30 pts), and Location/Relocation (10 pts) + Nice-to-haves (5 pts).

• Pipeline Step 4: Multiplies base match score by the behavioral signals envelope (0.1x to 1.0x).

• Pipeline Step 5: Sorts descending by rounded score. Ties are broken deterministically by Candidate ID in ascending order.

Explainability & Data Validation

- **Hallucination-Free Dynamic Reasoning & Verification:**

• Deterministic Explainability: Rather than relying on expensive, black-box LLMs that are prone to hallucinating skills, our script generates justifications using a rules-based natural language compiler.

• Factual Stitching: The reasoning dynamically references the candidate's exact years of experience, current title, matched skills (e.g. FAISS, Milvus), location alignment, and notice period constraints.

• Anti-Pattern Hashing: Deterministically hashes candidate IDs to vary sentence templates, introductory verbs, and connectors. This creates highly distinct, non-repetitive recruiter justifications across all 100 ranks.

• Data Integrity Check: The validation checks automatically catch experience inflation, dates out-of-bounds, and fake expert skills, routing anomalous profiles away from the ranking pool.

End-to-End Workflow

System Execution & Shortlist Generation Workflow:

- 1. Input Stream: Pipeline receives candidates JSONL dataset and job description parameters.
- 2. Honeypot Screening: Candidate profiles undergo chronological checks, instantly discarding 67 invalid profiles.
- 3. Feature Extraction: Candidate history, skills, location, and platform activity signals are loaded.
- 4. Matrix Evaluation: Calculates matching scores and multiplies them by the availability signals envelope.
- 5. Shortlisting & Sorting: Selects the top 100 candidates based on score and deterministic tie-breaking.
- 6. Explanation Compiler: Generates customized justifications for each of the top 100 entries.
- 7. CSV Output: Saves the final validated shortlist in the required format.

System Architecture

Architecture Components & Design Decisions:

• Ingestion Module: Memory-safe parser that streams candidates.jsonl line-by-line. Scalable to millions of records under the 16GB RAM constraint.

• Anomaly Interceptor: Screens candidate profiles against date overlaps, experience inflation, and zero-duration expert skills. Removes bad data before scoring.

• Deterministic Scorer: Standardized python scoring blocks executing base relevance matrix. Fast, highly reproducible, and 100% CPU-friendly.

• Signal Multiplier: Applies availability envelopes based on response rate, notice period, and active dates.

• Explainability Engine: A rule-based compiler that dynamically builds recruiting summaries without calling external LLM APIs.

Results & Performance

- ## Shortlist Output & Performance Results:

• Rank 1 Candidate: Anika Pillai (CAND_0007009) | Title: Recommendation Systems Engineer | Experience: 7.9 Years | Location: Noida, Uttar Pradesh, India | Matched Skills: Weaviate, OpenSearch | Notice Period: 30 Days | Recruiter Response: 62%

• Total Honeypot Rate: exactly 0.0% in the top 100, fully satisfying the Stage 3 check.

• Total Execution Runtime: ~12 seconds for the entire 100,000 candidate pool (Limit: 5 minutes wall-clock).

• Memory Consumption: < 200MB RAM, allowing it to run smoothly on any standard 16GB CPU machine.

• Network & API Dependency: 100% offline. Free from external API outages, latency spikes, or cost inflation.

Technologies Used

- ## Core Technology Choices & Rationale:

• Python Standard Library (json, csv, argparse, datetime): Used for the core ranking script. Guarantees zero installation overhead, rapid startup, and 100% compliance in offline sandboxes.

• Streamlit: Selected to build the sandbox web GUI, allowing recruiters to dynamically adjust matching weights, inspect profiles, and audit honeypot alerts in real-time.

• fpdf2: Used to programmatically render presentation PDFs for local submission preparation.

• Git & GitHub: Used to maintain a clean, collaborative codebase and track commit history and reproducibility.

Submission Assets

Key Submission Assets & Deliverables:

- GitHub Code Repository: https://github.com/mauliwanave/The-Data-AI-Challenge
- HuggingFace Sandbox / App Space: https://huggingface.co/spaces/mauliwanave/redrob-ranker
- Ranked Candidate Output: team_minipolling.csv (Format-validated containing the top 100 candidates)
- Approach Presentation PDF: approach_presentation.pdf (This slide deck converted to PDF)

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THANK YOU